

Seven QGIS Exercises, Step by Step

A workbook you can finish on your own, without the lecture

Every menu path, button and tick box in this workbook was read off a running copy of QGIS, not written from memory. Where a label looks oddly long, that is because the real label is oddly long. Type what you see here and you will find it on screen.

Each exercise has the same shape. **You will build** says what you get at the end, because every exercise produces one column or one file, never just a picture. **Files** names what to open. Then the numbered steps. **Check before moving on** gives you a number to compare against, so you find out immediately if something went wrong rather than three steps later. **If it goes wrong** lists the symptoms people actually hit, with the cause. **Why this matters** is the economics.

Work in order. Exercise 7 needs the map you build in Exercise 3, and Exercises 4 and 5 assume you have met the coordinate system problem in Exercise 1.

Before you start

1. Install **QGIS 3.x LTR** from qgis.org. Take the LTR (Long Term Release), not the newest one. The standard install is enough. You do not need a single plugin for this workbook.
2. Download the practice files from billytulungen.com/lab, the *Practice files* button. Unzip the folder somewhere you can find again, for example your Documents folder. **Do not work inside the zip file**. QGIS can sometimes open a file from inside a zip and then fail in confusing ways later.
3. Open QGIS. You get an empty project.
4. Turn on the **Processing Toolbox**: press **Ctrl+Alt+T**. A panel appears on the right with a search box at the top. Almost every tool in this workbook lives there, and you find it by typing its name rather than hunting through menus.
5. Look at the **bottom right corner** of the window. There is an EPSG code there, most likely **EPSG:4326**. Remember where that corner is. You will come back to it in Exercise 1, and it is not what most people think it is.

A note on saving. **Project > Save** writes a **.qgz** file that remembers which layers you loaded, their colours and their order. It does **not** contain the data. If you move the **.qgz** to another computer without the data folder, it opens empty. That is normal, not a broken file.

Exercise 1 Open a layer and check its CRS

You will build. Nothing yet. This one builds a habit: asking two questions of every spatial file anyone ever hands you.

Files. kabupaten.gpkg

1. Menu **Layer > Add Layer > Add Vector Layer...** A dialog opens.
2. Beside *Vector Dataset(s)*, click the ... button and pick kabupaten.gpkg from your unzipped folder. Click *Add*, then *Close*.
3. Twelve district polygons appear on the canvas. In the **Layers panel** on the left, the layer is called kabupaten. If you see nothing, right-click the layer and choose **Zoom to Layer(s)**.
4. Right-click the layer, choose **Properties...**, and click the **Information** tab on the left of the dialog that opens.
5. Scroll to the block headed *Coordinate Reference System (CRS)*. Read three lines: *Name* says *WGS 84*, *Units* says *degrees*, and the EPSG code is **EPSG:4326**. Close the dialog.
6. Now look at the bottom right corner of the main window again. That code is the **project** CRS. The one you just read in Properties is the **layer** CRS. They are two different settings and they can disagree without any warning.
7. Right-click the layer and choose **Open Attribute Table**, or just press **F6**. A table opens. Every row is one polygon on the map. Click a row and watch that district highlight.
8. Count the rows. The bar at the top of the table tells you.
9. Close the table. Choose **View > Identify Features (Ctrl+Shift+I)**, then click any district on the map. Its attributes appear in a panel. Press **Ctrl+Shift+I** again to switch the tool off, otherwise every later click keeps opening that panel.

Check before moving on. 12 rows in the attribute table. Layer CRS **EPSG:4326**, units *degrees*. Columns include kode_kab, nama_kab, luas_km2 and kemiskinan.

If it goes wrong. *The map is blank.* The canvas is zoomed somewhere else: right-click the layer, **Zoom to Layer(s)**. *Add stays grey.* You selected the folder instead of the file. *Every click opens the Identify panel.* The Identify tool is still active; press **Ctrl+Shift+I** to turn it off.

Why this matters. *Degrees* is the important word. A degree is a coordinate, not a length. One degree of latitude is about 110.6 km everywhere, but one degree of longitude is 111.3 km at the equator and only 55.8 km at latitude 60. Any distance or area you compute while the layer is in degrees is meaningless, and QGIS will not stop you. That is why Exercise 4 begins by reprojecting.

Exercise 2 Join a statistics table to the map

You will build. A district layer that carries poverty, population and consumption. This is `merge` in Stata, done through geometry instead of a `do-file`.

Files. `kabupaten_polos.gpkg` and `statistik_kabupaten.csv`

Start from a clean project: `Project > New`, and do not save the old one.

1. `Layer > Add Layer > Add Vector Layer...`, pick `kabupaten_polos.gpkg`, Add, Close. The same twelve polygons appear, but this copy carries only three columns: `kode_kab`, `nama_kab`, `luas_km2`. Press `F6` to confirm that.
2. Note that the layer appears in the Layers panel as `kabupaten`, not as `kabupaten_polos`. The file and the layer inside it have different names. That is normal for a GeoPackage.
3. `Layer > Add Layer > Add Delimited Text Layer...` A large dialog opens.
4. Beside *File name*, click `...` and pick `statistik_kabupaten.csv`. A preview table appears at the bottom under *Sample Data*.
5. Find the section headed **File Format**. *CSV (comma separated values)* should already be selected. Leave it.
6. Find the section headed **Record and Fields Options**. It may be collapsed into a single grey bar; if so, **click the bar to open it**. Inside, *First record has field names* is ticked. Leave that. Now **untick Detect field types**.
7. Find the section headed **Geometry Definition**. It is usually collapsed too. Click it open and choose **No geometry (attribute only table)**. The CSV has no coordinates, so this is correct.
8. Click *Add*, then *Close*. A new entry appears in the Layers panel with no map symbol beside it, because it is a table, not a map layer. Press `F6` on it: 12 rows, with a column called `kode`.
9. Now the join. Right-click the **polygon** layer (`kabupaten`), choose `Properties...`, then the **Joins** tab on the left.
10. Press the **+** button at the bottom left of that panel. The *Add Vector Join* dialog opens.
11. Set *Join layer* to the CSV table. Set *Join field* to `kode`, which is the key on the CSV side. Set *Target field* to `kode_kab`, which is the key on the map side. **The key has a different name on each side. That is deliberate, and it is what real data looks like.**
12. Tick *Custom field name prefix* and clear the box, or type something short. Without this, every joined column arrives with a long prefix stuck to its name.
13. Click OK, then OK again to close Properties.
14. Press `F6` on the polygon layer. The statistics columns are now there.

Check before moving on. All 12 rows carry values. No empty cells. If you see 12 rows of blanks in the joined columns, the join matched nothing.

If it goes wrong. *Every joined cell is empty.* You left *Detect field types* ticked at step 6. QGIS then read `kode` as a number while the map holds `kode_kab` as text, and a number never equals a string. Remove the join, remove the CSV, and load it again with that box unticked. *I cannot find Detect field types.* It is inside *Record and Fields Options*, which opens only when you click the grey bar. *Add* is grey in the delimited text dialog. You have not chosen *No geometry (attribute only table)* yet.

Now do it wrong on purpose. Remove the join, load the CSV again with *Detect field types* **ticked**, and join again. You get twelve empty cells and **no error message of any kind**. The layer still draws, the attribute table still opens, and a regression run on those columns would still produce coefficients.

Why this matters. A join that fails halfway is the most expensive silent error in applied work. Your sample changes, the regression still runs, and the rows that dropped are never random: they are the districts that split, merged or were renamed, and those differ from the rest in exactly the ways you are studying. Count your rows before the join, after the join, and how many matched. Write the three numbers down.

Exercise 3 *Make a thematic map*

You will build. A map of poverty, and proof that the classification you pick changes what a reader concludes. Save the project at the end: Exercise 7 starts from it.

Files. kabupaten.gpkg

Start a new project. Load kabupaten.gpkg the way you did in Exercise 1.

1. Right-click the layer, *Properties...*, then the **Symbology** tab.
2. At the very top of that panel is a dropdown currently reading *Single Symbol*. Change it to **Graduated**. The panel below changes completely.
3. Set *Value* to kemiskinan. That is the poverty rate, in percent.
4. Pick a *Color ramp*. Choose a single-hue ramp such as Blues or Greens. **Do not pick Spectral or any rainbow**. A rainbow has no natural order, so a reader cannot tell which end is high without checking the legend every time, and readers with red-green colour blindness cannot read it at all.
5. Set *Classes* to 5.
6. Choose a *Mode*. Start with *Equal Interval*.
7. Press **Classify**. **Nothing happens until you press it**. The list of classes fills in with break values.
8. Click *Apply*. The map colours in. Leave the dialog open.
9. Change *Mode* to *Equal Count (Quantile)*, press *Classify* again, then *Apply*. Watch the map change.
10. Change *Mode* to *Natural Breaks (Jenks)*, *Classify*, *Apply*. Watch it change again.
11. Click OK. Then *Project > Save As...* and save the project somewhere you will find it. Exercise 7 continues from here.

Check before moving on. The twelve poverty values run from 21.2 to 48.0. With five classes you should get these break values. *Equal Interval*: 26.6, 31.9, 37.3, 42.7. *Equal Count (Quantile)*: 22.2, 23.8, 26.4, 30.9. *Natural Breaks (Jenks)*: 22.7, 25.5, 28.6, 36.7.

If it goes wrong. *The class list is empty and the map does not change*. You did not press *Classify*. *I cannot find Quantile in the dropdown*. It is written *Equal Count (Quantile)*, and Jenks is written *Natural Breaks (Jenks)*.

Look at what just happened. Same column, same twelve districts, same five classes, same colours. Only the mode changed, and the three maps tell three different stories.

Count how many districts land in each class and you can see why. Under *Equal Interval* the classes are all 5.4 wide, but the districts fall 7, 3, 1, 0, 1: over half of them share the lowest class and one class is completely empty. The map looks calm. Under *Equal Count (Quantile)* the districts fall 3,

2, 2, 2, 3, but the class widths are 1.0, 1.6, 2.6, 4.5 and 17.1. Two districts 0.9 apart get different colours while two districts 16.5 apart share one. The map looks alarming. Under *Natural Breaks (Jenks)* the cuts land in the gaps where no district sits.

Why this matters. Each mode is honest about something different. Equal Interval is honest about the value scale and hides the shape of the distribution. Quantile is honest about rank and hides magnitude. Natural Breaks is honest about the structure of *this* dataset, which means it is not comparable with any other. So if you map 2015 and 2020 side by side and leave the mode on automatic, the two maps use different breaks and the same colour means different numbers. For a series of maps you must fix the breaks by hand. And whichever you pick, name the mode and the number of classes in your figure caption. It is a research choice, not a formatting choice.

Exercise 4 Buffer and spatial join

You will build. A column called `within5km`, equal to 1 for villages within five kilometres of a main road and 0 for the rest. That column is a treatment or exposure variable.

Files. `desa.gpkg` and `jalan.gpkg`

Start a new project and load both files. You get 900 village points and 4 road lines, both in [EPSG:4326](#), which means degrees, which cannot measure anything. Fix that first.

Part A: reproject both layers to metres

1. Press **Ctrl+Alt+T** for the Processing Toolbox. In its search box type *reproject*.
2. Under the group **Vector general** you will see **Reproject layer**. Double-click it.
3. Set *Input layer* to `desa`.
4. Go to *Target CRS*. **Do not use the dropdown**. That dropdown lists only the coordinate systems you have used before, so on a fresh install it is nearly empty and [EPSG:32749](#) is not in it. This is the single most common place people get stuck.
5. To the right of the dropdown is a small **globe button**. Click it. The coordinate system chooser opens.
6. At the top of the chooser is a *Filter* box. Type `32749`.
7. The upper panel, *Recently Used*, stays empty. That is expected. Look at the **lower** panel, *Predefined Coordinate Reference Systems*, under *Projected* then *Universal Transverse Mercator (UTM)*. The line reads *WGS 84 / UTM zone 49S* with [EPSG:32749](#) beside it. Click it, then OK.
8. *Target CRS* now reads [EPSG:32749 - WGS 84 / UTM zone 49S](#). Press **Run**. A new layer called *Reprojected* appears. Rename it so you know which is which.
9. Repeat steps 2 to 8 for `jalan`.

If it goes wrong. **Do not change the CRS in** [Properties](#) > [Source](#). That box relabels the layer, it does not convert anything. The coordinates stay as degrees while the layer claims to be in metres. Everything then collapses to a single speck near Antarctica, about 9,800 km from where it should be, and the symptom you see is an empty map or all points stacked on one spot. *Reproject layer* recalculates the coordinates; *Set CRS* lies to QGIS about what it is holding.

Part B: draw the five-kilometre zone

10. In the Toolbox search box type *buffer*. Under **Vector geometry** open **Buffer**.
11. *Input layer* is the **reprojected roads**, not the original.
12. *Distance*: type **5000**. Look at the small unit box beside it. It should read *meters*. **If it reads degrees, stop** and go back to Part A, because your input layer is the unprojected one.
13. Tick **Dissolve result**. Without this you get four overlapping buffers and villages near two roads get counted twice.
14. Press Run. A layer called *Buffered* appears: one continuous band along the roads.

Part C: find out which villages fall inside it

15. In the Toolbox type *join attributes*. Under **Vector general** open **Join attributes by location**.
16. *Join to features in*: the **reprojected villages**. This is the layer that gets the new columns.
17. Under *Features they (geometric predicate)*, tick **intersect**. Leave the other six alone.
18. *By comparing to*: the *Buffered* layer.
19. *Join type*: leave it on *Take attributes of the first matching feature only (one-to-one)*.
20. **Leave Discard records which could not be joined unticked**. This is the whole point. Ticked, you get back only the villages near a road and you have thrown away your control group. Unticked, you get all 900 back, and the ones near a road carry the extra columns.
21. Press Run. A layer called *Joined layer* appears. Press **F6**.
22. Open the Field Calculator: the abacus icon in the attribute table toolbar, or **Ctrl+I** with the table open. Tick *Create a new field*, name it *within5km*, set the type to *Integer*, and enter this expression:

```
nama_ruas IS NOT NULL
```

Press OK. The table is now in edit mode. Press **Ctrl+E** to toggle editing off, and choose *Save* when it asks.

Check before moving on. 900 rows in the joined layer, of which **147** have a value in *nama_ruas* and therefore *within5km* equal to 1. If you get 147 rows in total instead of 900, you ticked *Discard records*. If you get more than 900, you forgot *Dissolve result*.

Why this matters. Notice what the buffer actually produced: not a picture of circles, but a column. A GIS operation should hand you a variable, never a smaller sample. The moment you let it filter your data instead of labelling it, you have deleted your control group and no amount of clever econometrics afterwards will bring it back. The same forty-mile buffer, drawn around American railways in 1890, is Figure I of a Quarterly Journal of Economics paper. The tool is not what makes it research. The question is.

Exercise 5 Zonal statistics and export to CSV

You will build. A CSV with one row per district and a column holding the average night-time light inside it. This is the moment a satellite image becomes a variable in a regression.

Files. *nightlights.asc* and *kabupaten.gpkg*

Start a new project.

1. Layer > Add Layer > Add Vector Layer..., pick kabupaten.gpkg.
2. Layer > Add Layer > Add Raster Layer..., pick nightlights.asc. A grey grid appears. Drag it **below** the districts in the Layers panel so you can see both.
3. Press **Ctrl+Alt+T**, and in the Toolbox search box type *zonal*.
4. Under the group **Raster analysis** open **Zonal statistics**. Take care: there is an older tool with a similar name. The one you want is in Raster analysis and its first box is called *Input layer*.
5. *Input layer*: kabupaten. **This is the polygons, not the raster.** The name misleads almost everyone the first time.
6. *Raster layer*: nightlights.
7. *Raster band*: leave it on Band 1. This raster has only one.
8. *Output column prefix*: type something you will recognise later, for example *ntl_*. Every new column will start with it.
9. Open *Statistics to calculate*. The list starts *Count, Sum, Mean*. Tick those three.
10. Press Run. A layer called *Zonal Statistics* appears. Press **F6** on it: three new columns at the right end of the table.
11. Right-click that layer, choose **Export > Save Features As...**
12. Set *Format* to **Comma Separated Value [CSV]**. Choose a file name and folder. Press OK.
13. Open the CSV in a spreadsheet or a text editor. One row per district, with the district code, the mean light and the cell count.

Check before moving on. 12 rows out, one per district, none of them empty. Every district should have a count of several hundred cells or more. As a reference point, one district in this dataset covers 5,909 cells, with a mean of 3.299 and a sum of 19,492.7 over an area of 29,096 square kilometres.

If it goes wrong. *The result is empty or every value is NULL.* Check the two CRS labels in the dialog. If the raster and the polygons are in different coordinate systems, the tool looks for cells where there are none. *I ticked the wrong boxes.* The order in the list is Count, Sum, Mean, not mean, sum, count. *Only some districts have values.* Some polygons sit partly outside the raster; check the count column to see which.

Always ask for *Count* alongside *Mean*, even when you do not think you need it. A small district may contain only a dozen cells, and a mean resting on a dozen cells is a very different number from a mean resting on six thousand. Without the count column your reader cannot tell the two apart, and neither can you.

Why this matters. This is the step that turns an image archive into panel data. Run it again on *elevasi.asc*, which is in the same folder, and you have a second variable. Run it on next year's lights and stack the two results, and you have a district-by-year panel. Everything after that is econometrics you already know. One caution for real work: *mean* is safe, but *sum* depends on how many cells happen to fall inside a polygon, so it is not a physical total unless you weight it by cell area.

Exercise 6 Make a point layer from a CSV

You will build. A map layer built from nothing but a table of coordinates. This is how most survey data becomes spatial.

Files. `village_coordinates.csv`

1. Open the CSV in a text editor first and look at the header row. The first two columns are called X and Y. **X is longitude and Y is latitude**, in that order.
2. In QGIS, `Layer > Add Layer > Add Delimited Text Layer...`
3. Beside *File name* click `...` and pick `village_coordinates.csv`.
4. In **File Format**, leave *CSV (comma separated values)* selected.
5. Open the **Geometry Definition** section. **It stays collapsed even after you choose the file**, so you have to click the grey bar yourself. This is the single most common place people stall in this exercise.
6. Choose **Point coordinates**.
7. Set *X field* to X and *Y field* to Y. QGIS usually guesses these correctly, but check them rather than trusting the guess.
8. Set *Geometry CRS* to `EPSG:4326`, because these coordinates are degrees. Use the globe button and the *Filter* box if the code is not in the dropdown, exactly as in Exercise 4.
9. The *Add* button stays grey until the CRS is set. Once it is live, click *Add*, then *Close*.
10. The points appear. Check they land over the districts from the earlier exercises and not somewhere in the ocean.
11. Right-click the layer, `Export > Save Features As...`, format *GeoPackage*, and save it. Until you do this the layer exists only inside the project.

Check before moving on. 900 points, in roughly the same place as the district polygons.

If it goes wrong. *All the points are in the Atlantic Ocean off West Africa.* You swapped X and Y, or the CRS is wrong. Longitude zero and latitude zero meet at exactly that spot, which is why broken coordinate data always lands there. Suspect your columns before you suspect Indonesian geography. *Add stays grey.* *Geometry CRS* is not set yet. *I cannot find X field.* The *Geometry Definition* section is still collapsed.

Why this matters. The reason people swap X and Y is that we say *latitude, longitude* in conversation while every piece of software wants X, Y, which means *longitude, latitude*. The spoken order is the reverse of the machine order. Once you have points, you can attach them to anything else that has a location: rainfall grids, road networks, administrative boundaries.

Exercise 7 Print Layout for a figure you can publish

You will build. A PDF of your poverty map that could go into a thesis without apology.

Files. The project you saved at the end of Exercise 3

1. Open that project. **Project > New Print Layout...** (Ctrl+P). Name it after the figure number it will carry, for example *Figure 1*. A separate window opens: a blank page.
2. **Add Item > Add Map**. Your cursor becomes a cross. Drag a rectangle across most of the page. Your map appears inside it.
3. **Add Item > Add Legend**, then click once on the page. The legend appears with your five classes in it.
4. In the panel on the right, under *Item Properties*, untick *Auto update*, then double-click each class label to rename it. Change kemiskinan to something a reader understands, such as *Poverty rate (%)*.
5. **Add Item > Add Scale Bar**, click on the page, and drag it to a corner.
6. **Add Item > Add North Arrow** only if the map is rotated. If north is straight up, as it is here, a north arrow adds nothing and you may leave it out.
7. **Add Item > Add Label**, drag a box, and type a real title: what, where, and when. Add a second label for the source line and the classification method, for example *Source: synthetic practice data. Quantile classification, 5 classes*.
8. Look at the **Items** panel on the right. Every element you added is a separate row there, and you can select, move and edit each one at any time. A screenshot of the map canvas gives you none of this.
9. **Layout > Export as PDF...** For a raster image instead, **Layout > Export as Image...** at 300 dpi.

Check before moving on. Your PDF has a title, a legend with readable labels, a scale bar and a source line, and any district with no data looks different from a district whose value is zero.

Why this matters. A map for publication is not a screenshot. The checklist is short: the title says what, where and when; the legend uses plain units rather than raw column names; the source and year are given; the classification method and number of classes are stated; the colours survive being printed in black and white; and areas with no data are visibly different from areas with a value of zero. That last one is violated most often and matters most. Colouring a missing district the same as the lowest class tells your reader it is the poorest, when the truth is that you do not know.

When you have finished all seven

Two extra files are in the bundle for exactly this moment.

`elevasi.asc` is a second raster on the same grid. Run Exercise 5 again on it and you have a second control variable, which is the normal way geographic controls get built.

`podes_sample.csv` joins to `desa.gpkg` through the column `iddesa`, matching against `kode_desa` on the village side. **It is broken on purpose.** It has 888 rows, of which 863 match and **25 do not**. Your job is to find those 25 and work out what they have in common. This is the closest thing in the bundle to real data, and it is the skill that separates work a referee trusts from work a referee does not.

Then do the whole thing once on a province you actually care about. Take district boundaries from BPS and one indicator you find interesting. Join them on the code, count your rows before and after, and find out who dropped. Map it in five classes and be ready to say why that classification and not another. Build one distance variable. Export to CSV and run a descriptive regression on that distance, with no causal reading.

Nobody will collect it. That is the point: you will only do it if you want to be able to do it.